Unit Test Report — Bank Management System

1. Executive Summary and Test Environment

This report presents the formal unit and integration test execution results for the Bank Management System. All 76 automated test cases were executed using Pytest 8.2.2 against an isolated SQLite test database with write-ahead logging (WAL) and foreign keys enabled. The test suite achieved a 100.0% pass rate (76 passed, 0 failed, 0 skipped) in 172.85 seconds.

2. Test Module Breakdown

Test Module File	Target Functional Modules	Total Tests	Passed	Failed	Pass Rate
tests/test_all.py	Authentication, RBAC, Accounts, Transactions, Transfers, Loans, Beneficiaries, Diagnostics	49	49	0	100.0%
tests/test_backup_recovery.py	Online SQLite Backup, Checksum Verification, Pre-restore Snapshots, Disaster Recovery	5	5	0	100.0%
tests/test_issue_13_15.py	Transaction History, Date and Type Filtering, Pagination, CSV Formula Sanitization	9	9	0	100.0%
tests/test_notifications.py	In-App Event Alerts, Unread Counters, State Updates, Zero Queue Latency	7	7	0	100.0%
tests/test_profile_settings.py	Profile Updates, Email Uniqueness, Password Policy Verification, Cross-User Isolation	6	6	0	100.0%

3. Comprehensive Unit Test Execution Log

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-01	tests/test_all.py	FR-002	`test_customer_registration_success`	User registration generates record and primary savings account	**Pass**
TC-02	tests/test_all.py	FR-002	`test_customer_registration_duplicate_username_email`	Duplicate username or email is rejected with HTTP 400	**Pass**
TC-03	tests/test_all.py	FR-002	`test_customer_registration_weak_password`	Passwords under 8 chars or missing complexity fail validation	**Pass**
TC-04	tests/test_all.py	FR-002	`test_login_success_and_session_persistence`	Valid credentials establish authenticated session with HttpOnly cookie	**Pass**
TC-05	tests/test_all.py	FR-002	`test_login_invalid_password`	Invalid credentials return generic HTTP 401 error	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-06	tests/test_all.py	FR-002	`test_login_brute_force_lockout`	5 consecutive bad logins trigger 15-minute account lockout	**Pass**
TC-07	tests/test_all.py	FR-002	`test_logout`	Logout destroys session and revokes authentication state	**Pass**
TC-08	tests/test_all.py	SR-004	`test_csrf_protection_on_mutating_endpoints`	Mutating POST/PUT without CSRF token returns HTTP 403 Forbidden	**Pass**
TC-09	tests/test_all.py	FR-003	`test_rbac_admin_endpoint_forbidden_for_customer`	Customer access to /api/admin returns HTTP 403 Forbidden	**Pass**
TC-10	tests/test_all.py	FR-003	`test_rbac_admin_can_list_users`	Admin can list system users and inspect account metadata	**Pass**
TC-11	tests/test_all.py	FR-003	`test_rbac_admin_cannot_demote_self`	Admin cannot revoke own administrative privileges	**Pass**
TC-12	tests/test_all.py	FR-003	`test_rbac_admin_cannot_deactivate_self`	Admin cannot deactivate own active user account	**Pass**
TC-13	tests/test_all.py	FR-003	`test_rbac_ownership_isolation_between_customers`	Customer A cannot access Customer B accounts or ledger	**Pass**
TC-14	tests/test_all.py	FR-003	`test_rbac_employee_can_access_any_customer_account`	Employee role can access any customer account for support	**Pass**
TC-15	tests/test_all.py	FR-004	`test_customer_can_open_savings_account_for_self`	Customer can open new savings account with unique 12-digit number	**Pass**
TC-16	tests/test_all.py	FR-004	`test_employee_account_creation_for_customer`	Employee can provision banking account on behalf of customer	**Pass**
TC-17	tests/test_all.py	FR-004	`test_account_status_update_and_closure_rules`	Account transitions between ACTIVE, FROZEN, and CLOSED correctly	**Pass**
TC-18	tests/test_all.py	FR-005	`test_deposit_valid_amount_updates_balance`	Valid deposit increases balance in paise and updates ledger	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-19	tests/test_all.py	FR-005	`test_deposit_invalid_negative_or_zero_amount`	Negative and zero deposit amounts return HTTP 400 Bad Request	**Pass**
TC-20	tests/test_all.py	FR-005	`test_deposit_to_frozen_or_closed_account_rejected`	Deposits to frozen or closed accounts are rejected	**Pass**
TC-21	tests/test_all.py	FR-006	`test_withdraw_valid_amount_within_balance`	Withdrawal within balance debits account and updates ledger	**Pass**
TC-22	tests/test_all.py	FR-006	`test_withdraw_exceeding_balance_rejected`	Overdraft attempt returns HTTP 400 Insufficient funds	**Pass**
TC-23	tests/test_all.py	FR-006	`test_withdraw_from_other_customer_account_forbidden`	Customer cannot withdraw from another user account	**Pass**
TC-24	tests/test_all.py	FR-007	`test_transaction_ledger_records_and_retrieval`	Ledger entry records reference ID, timestamp, and resulting balance	**Pass**
TC-25	tests/test_all.py	NFR-022	`test_audit_log_entries_created`	Audit events generated on logins, financial operations, and transfers	**Pass**
TC-26	tests/test_all.py	FR-008	`test_transfer_between_own_accounts_success`	Transfer between customer own accounts updates both balances	**Pass**
TC-27	tests/test_all.py	FR-008	`test_transfer_to_another_customer_account_and_transfer_history`	Inter-customer transfer updates sender and receiver records	**Pass**
TC-28	tests/test_all.py	FR-008	`test_transfer_insufficient_funds_rejected_and_atomic`	Transfer exceeding balance aborts without partial debit	**Pass**
TC-29	tests/test_all.py	FR-008	`test_transfer_validation_errors`	Self-transfers and invalid destination accounts are rejected	**Pass**
TC-30	tests/test_all.py	FR-008	`test_transfer_frozen_or_closed_accounts_rejected`	Transfers involving frozen or closed accounts are blocked	**Pass**
TC-31	tests/test_all.py	FR-009	`test_atomic_transfer_rollback_on_mid_transaction_failure`	Transfer failure mid-transaction triggers full database rollback	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-32	tests/test_all.py	NFR-019	`test_issue_19_security_headers_and_cache_control`	Responses contain CSP, X-Frame-Options: DENY, Cache-Control	**Pass**
TC-33	tests/test_all.py	NFR-019	`test_issue_19_no_plaintext_passwords_in_db_or_api`	Database passwords stored strictly as salted bcrypt hashes	**Pass**
TC-34	tests/test_all.py	NFR-020	`test_issue_20_system_response_time_under_threshold`	Core endpoints respond well within latency benchmark	**Pass**
TC-35	tests/test_all.py	NFR-021	`test_health_check`	Health endpoint /api/health reports UP status and DB connectivity	**Pass**
TC-36	tests/test_all.py	NFR-021	`test_issue_21_health_and_readiness_diagnostics`	Diagnostic checks confirm database, disk, and memory availability	**Pass**
TC-37	tests/test_all.py	NFR-021	`test_issue_21_graceful_degradation_and_recovery_on_db_failure`	System handles database unavailability returning HTTP 503	**Pass**
TC-38	tests/test_all.py	NFR-021	`test_issue_21_teardown_rollback_and_fault_isolation`	Request teardown releases SQLite locks and prevents leaks	**Pass**
TC-39	tests/test_all.py	NFR-021	`test_issue_21_sustained_load_availability_and_consistency`	Rapid sequential requests maintain strict ledger consistency	**Pass**
TC-40	tests/test_all.py	FR-010	`test_issue_10_customer_submit_loan_application_success`	Customer submits valid loan and receives unique reference ID	**Pass**
TC-41	tests/test_all.py	FR-010	`test_issue_10_loan_application_validation_errors`	Missing fields or invalid amounts reject loan application	**Pass**
TC-42	tests/test_all.py	FR-011	`test_issue_11_loan_review_queue_and_ownership_isolation`	Employees view review queue; customer access is forbidden	**Pass**
TC-43	tests/test_all.py	FR-012	`test_issue_12_loan_approval_and_rejection_workflow`	Loan approval and rejection workflow; rejection mandates reason	**Pass**
TC-44	tests/test_all.py	FR-017	`test_issue_17_add_list_update_delete_beneficiary_success`	CRUD operations on beneficiaries succeed for valid accounts	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-45	tests/test_all.py	FR-017	`test_issue_17_beneficiary_validation_and_duplicate_prevention`	Duplicate beneficiary account for same customer is rejected	**Pass**
TC-46	tests/test_all.py	FR-017	`test_issue_17_beneficiary_ownership_isolation_and_transfer_integration`	Customer cannot access another customer beneficiaries	**Pass**
TC-47	tests/test_all.py	NFR-022	`test_issue_22_audit_logs_rbac_and_comprehensive_event_recording`	Audit log records actor, IP, timestamp, action, and details	**Pass**
TC-48	tests/test_all.py	NFR-022	`test_issue_22_audit_logs_filtering_pagination_and_validation`	Admin filters audit trail by actor, action, and date range	**Pass**
TC-49	tests/test_all.py	NFR-022	`test_issue_22_audit_logs_csv_export_and_immutability`	Audit logs export as CSV; records cannot be modified via API	**Pass**
TC-50	tests/test_backup_recovery.py	NFR-024	`test_admin_create_backup_success`	Admin creates online SQLite backup with SHA-256 checksum	**Pass**
TC-51	tests/test_backup_recovery.py	NFR-024	`test_verify_backup_integrity`	Integrity check executes PRAGMA integrity check and verifies hash	**Pass**
TC-52	tests/test_backup_recovery.py	NFR-024	`test_rbac_protects_backup_endpoints`	Non-admin roles are rejected from backup endpoints with HTTP 403	**Pass**
TC-53	tests/test_backup_recovery.py	NFR-024	`test_end_to_end_disaster_recovery_restoration`	System creates pre-restore snapshot and restores from backup	**Pass**
TC-54	tests/test_backup_recovery.py	NFR-024	`test_restore_rejects_corrupted_backup`	Restoration aborts when backup SHA-256 checksum does not match	**Pass**
TC-55	tests/test_issue_13_15.py	FR-013	`test_history_has_pagination_metadata`	History response contains total count, total pages, current page	**Pass**
TC-56	tests/test_issue_13_15.py	FR-014	`test_type_filter_sort_and_pagination`	Filters ledger records by DEPOSIT, WITHDRAWAL, TRANSFER	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-57	tests/test_issue_13_15.py	FR-014	`test_search_filter`	Filters transactions by reference ID or description keyword	**Pass**
TC-58	tests/test_issue_13_15.py	FR-014	`test_inclusive_date_filter`	Filters transactions within inclusive start and end dates	**Pass**
TC-59	tests/test_issue_13_15.py	FR-014	`test_bad_filters_are_rejected`	Malformed dates and invalid page parameters return HTTP 400	**Pass**
TC-60	tests/test_issue_13_15.py	FR-013	`test_other_customers_history_is_forbidden`	Customer cannot query another user transaction history	**Pass**
TC-61	tests/test_issue_13_15.py	FR-015	`test_csv_export_respects_filters_and_attachment_headers`	CSV statement export sets attachment header and filter params	**Pass**
TC-62	tests/test_issue_13_15.py	SR-005	`test_csv_export_escapes_spreadsheet_formula_injection`	Prepends apostrophe to cells starting with =, +, -, @	**Pass**
TC-63	tests/test_issue_13_15.py	FR-015	`test_csv_export_includes_rows_after_first_hundred`	Streams full ledger without arbitrary 100-row truncations	**Pass**
TC-64	tests/test_notifications.py	FR-016	`test_deposit_triggers_notification`	Cash deposit generates synchronous in-app customer alert	**Pass**
TC-65	tests/test_notifications.py	FR-016	`test_withdrawal_triggers_notification`	Cash withdrawal generates synchronous in-app customer alert	**Pass**
TC-66	tests/test_notifications.py	FR-016	`test_transfer_triggers_dual_notifications`	Transfer creates debit alert for sender and credit for receiver	**Pass**
TC-67	tests/test_notifications.py	FR-016	`test_loan_lifecycle_notifications`	Application submission, approval, and rejection trigger alerts	**Pass**
TC-68	tests/test_notifications.py	FR-016	`test_mark_as_read_and_mark_all_read`	Updates read status and decrements unread count accurately	**Pass**
TC-69	tests/test_notifications.py	FR-016	`test_notification_ownership_isolation`	Customer cannot view or modify another customer alerts	**Pass**

ID	Module File	Req ID	Test Function Name	Test Objective and Assertions	Status
TC-70	tests/test_notifications.py	FR-016	`test_notification_delivery_latency_sla`	Alert committed in same transaction with zero queue latency	**Pass**
TC-71	tests/test_profile_settings.py	FR-018	`test_profile_requires_authentication`	Unauthenticated GET /api/profile returns HTTP 401 Unauthorized	**Pass**
TC-72	tests/test_profile_settings.py	FR-018	`test_profile_update_refreshes_me_and_persists`	Updating profile details persists across subsequent requests	**Pass**
TC-73	tests/test_profile_settings.py	FR-018	`test_profile_rejects_duplicate_email`	Email change to an existing email returns HTTP 400 error	**Pass**
TC-74	tests/test_profile_settings.py	FR-018	`test_profile_validation_and_csrf`	Malformed email formats and missing CSRF tokens are rejected	**Pass**
TC-75	tests/test_profile_settings.py	FR-018	`test_password_change_checks_current_password_and_policy`	Password change verifies old password and password policy	**Pass**
TC-76	tests/test_profile_settings.py	FR-018	`test_user_cannot_change_another_users_profile`	Cross-user profile modification attempt returns HTTP 403	**Pass**

4. Defect Log and Verification Outcome

All defects identified during earlier test iterations have been resolved:

- IT-03 Notification Delay: Resolved. Alerts now commit synchronously with database transactions.
- UT-12 Self-Transfer: Resolved. Server validates source_account_id != destination_account_id returning HTTP 400.
- IT-09 Loan Rejection Reason: Resolved. Mandatory rejection_reason field enforced and returned in notifications.
- ST-10 CSV Statement Export: Resolved. Timestamps standardized to ISO 8601; cells starting with =, +, -, @ sanitized.